

Analog IC Design ASSESSMENT 1

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EXPERIMENT 1: MOS DEVICE CHARACTERISTICS

1. NMOS Characteristics

OBJECTIVE

- To obtain the Drain Current (I_D) versus Gate-to-Source Voltage (V_{GS}) characteristics of the MOS transistor.
- To obtain the Drain Current (I_D) versus Drain-to-Source Voltage (V_{DS}) characteristics for various values of (V_{GS}) and determine the channel-length modulation parameter (λ)
- To obtain the Drain Current (I_D) versus Gate-to-Source Voltage (V_{GS}) characteristics for various values of Source-to-Body Voltage (V_{SB}) and study the body effect.

PROCEDURE

1. Open MobaXterm and create a new SSH session.
2. Connect to the Cadence server using the provided remote host, username, and password.
3. Launch the Cadence environment by entering `csd` and start Virtuoso.
4. Create a new library and attach it to the UMC-18 CMOS technology library.
5. Create a new schematic cell view.
6. Place an NMOS transistor (N_18_MM) from the UMC-18 CMOS library.
7. Place gnd, vdc sources, and other required components from the analogLib library.
8. nConnect the circuit using the wire tool and assign the transistor dimensions:
 - Channel Length (L) = 500 nm
 - Channel Width (W) = 10 μ m
9. Define the voltage sources:
 - V0 = V_{GS}
 - V1 = V_{DS}
10. Save and check the schematic for errors.
11. Launch ADE L (Analog Design Environment).
12. Load the UMC-18 CMOS Spectre model library and select the TT process corner.
13. Select DC Analysis and enable Save DC Operating Point.
14. Copy the design variables from the cell view.
15. Perform DC sweep and parametric analysis to obtain:

- (ID) vs. (VGS)
- (ID) vs. (VDS) for different values of (VGS)
- (ID) vs. (VGS) for different values of (VSB)

16. Simulate the circuit, plot the required characteristics, and record the results.

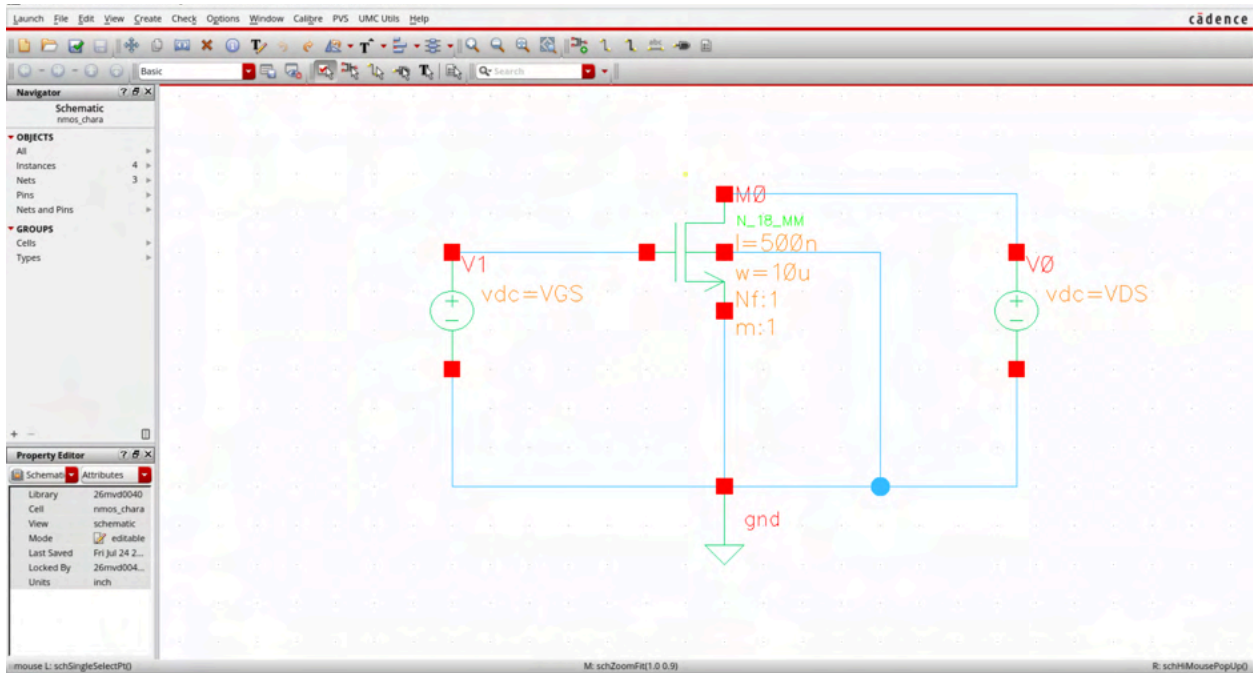


Figure 1: NMOS Characterization Circuit for ID V/s VGS and ID V/s VDS Analysis .

The circuit consists of an **NMOS transistor** configured for obtaining its **output characteristics (ID V/s VDS)**. The **source** and **body (bulk)** terminals are connected to **ground**, ensuring $V_{BS}=0$. The **gate** terminal is connected to a variable DC voltage source (**V1**) that provides the required **gate-to-source voltage (VGS)**, while the **drain** terminal is connected to another variable DC voltage source (**V0**) that provides the **drain-to-source voltage (VDS)**. During the DC sweep, VDS is varied over the desired range for different fixed values of VGS, and the corresponding drain current (ID) is measured. The resulting characteristics are used to study the cut-off, linear, and saturation regions of operation and to determine the channel length modulation parameter (λ).

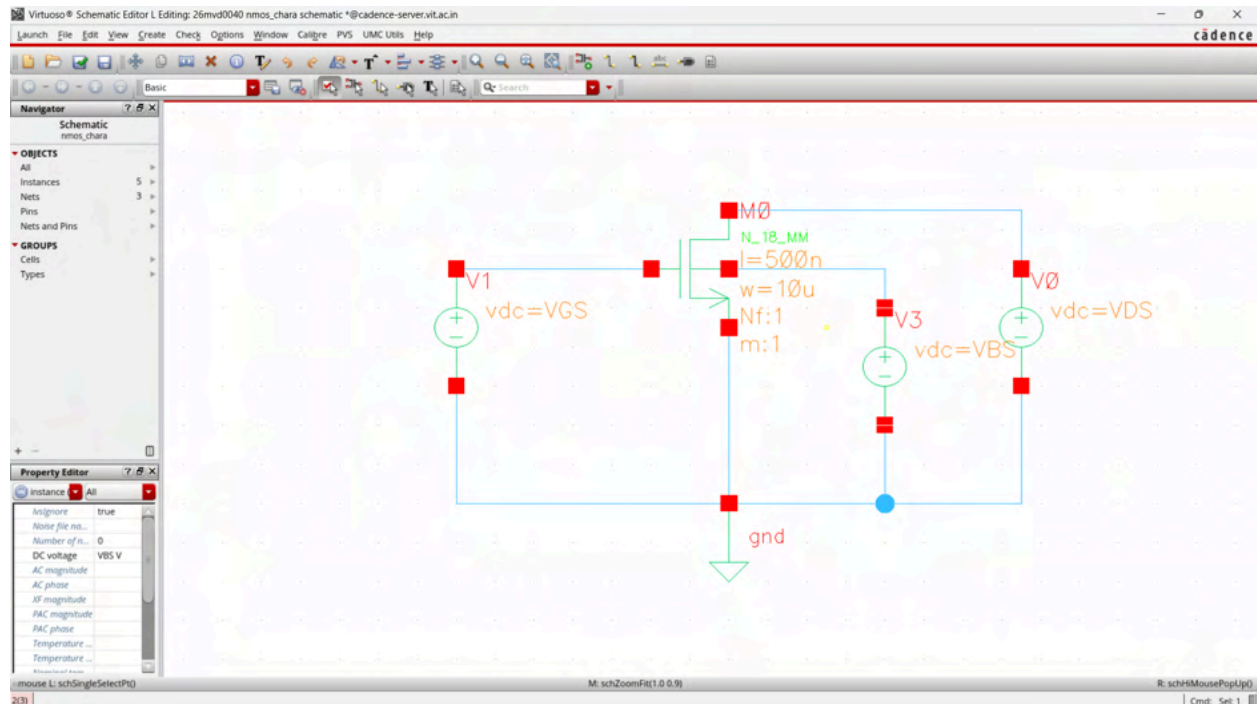


Figure 2: NMOS Test Circuit for Body Effect Analysis (I_D) Vs (V_{GS}) for Various (V_{BS})

The circuit consists of an **NMOS transistor** configured to study the **body effect** by varying the **body-to-source voltage (V_{BS})**. The **gate** is connected to a variable DC voltage source (**V1**) that provides the **gate-to-source voltage (V_{GS})**, while the **drain** is connected to another DC voltage source (**V0**) that provides the **drain-to-source voltage (V_{DS})**. A third DC voltage source (**V3**) is connected between the **body (bulk)** and **ground** to apply the desired **body bias (V_{BS})**. During simulation, (V_{GS}) is swept while (V_{DS}) is kept constant, and (V_{BS}) is varied parametrically. The corresponding **drain current (I_D)** is measured to obtain the **transfer characteristics (I_D – V_{GS})** for different body bias voltages, demonstrating the effect of body bias on the threshold voltage and drain current of the NMOS transistor.

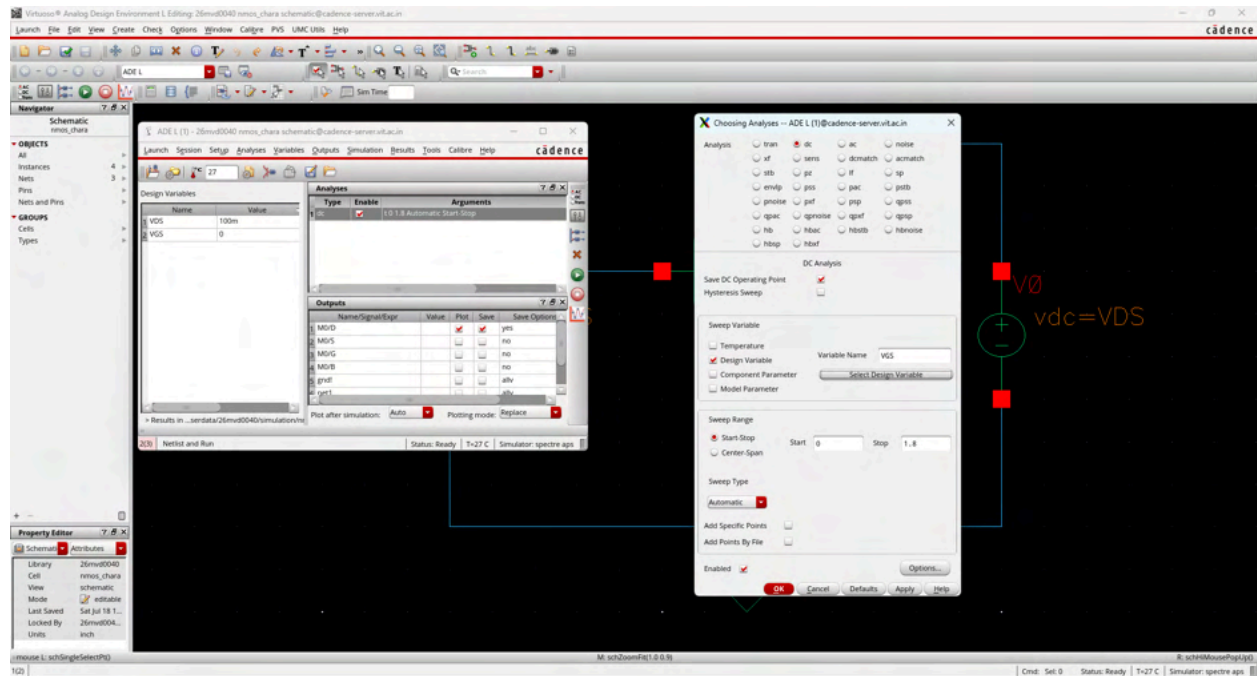


Figure 3: Configuration of DC Analysis for NMOS ID V/s VGS Simulation.

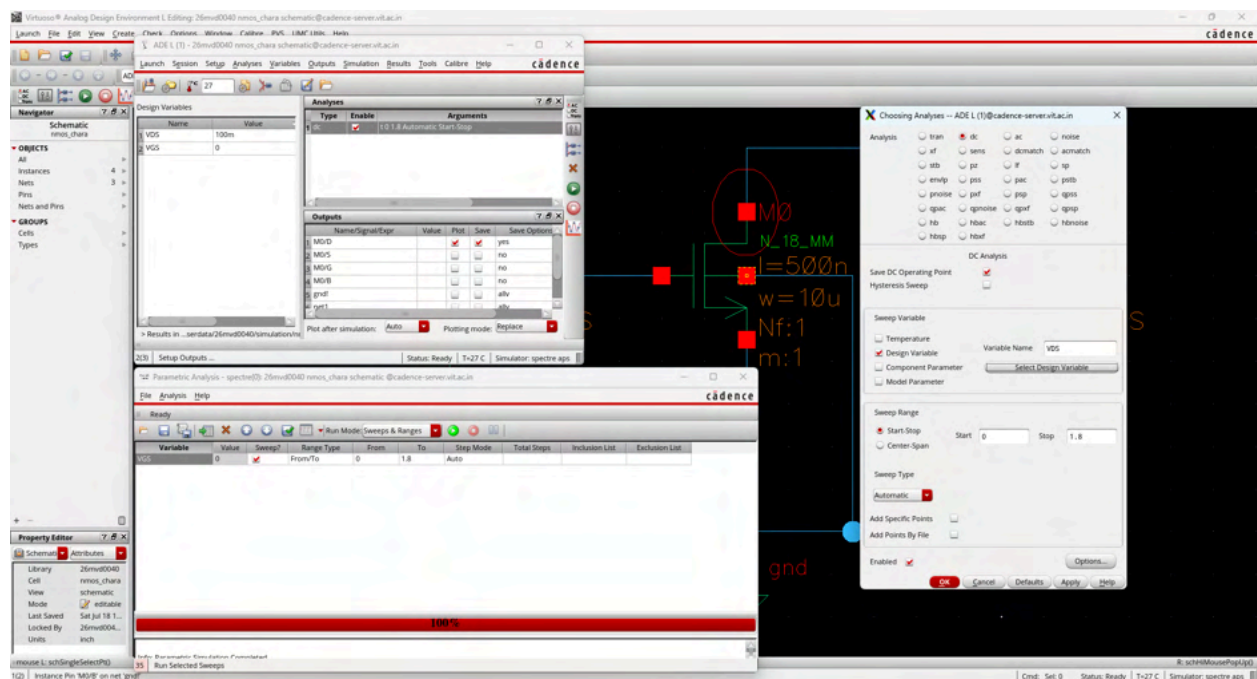


Figure 4: Configuration of DC Analysis for NMOS ID V/s VDS Simulation.

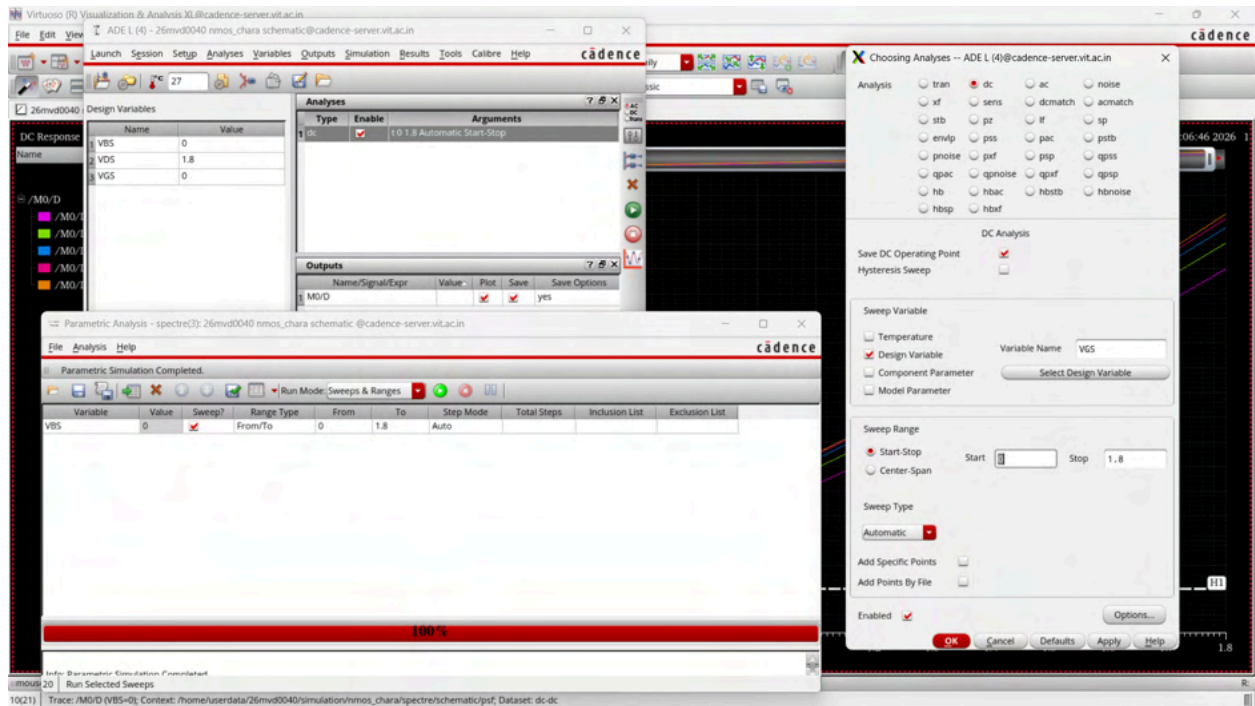


Figure 5: DC Parametric Sweep for ID V/s VGS Characteristics with Variable VBS.

OBSERVATIONS

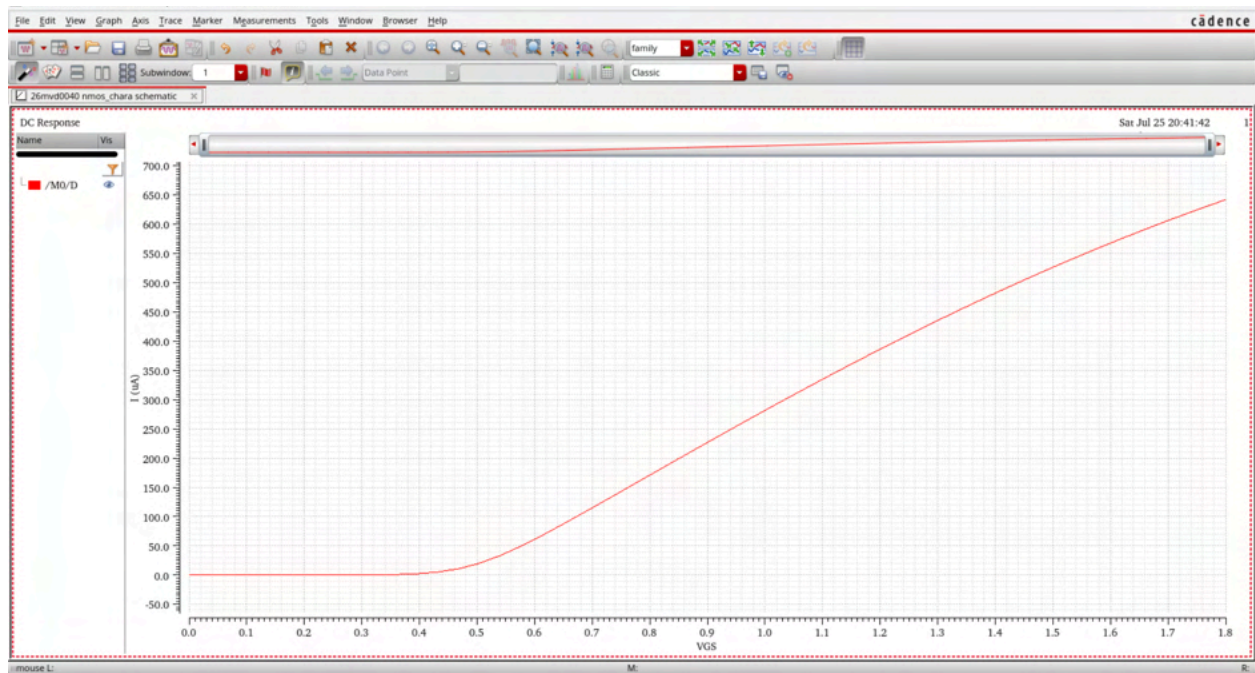


Figure 6: Transfer characteristics (ID) vs. (VGS) were obtained.

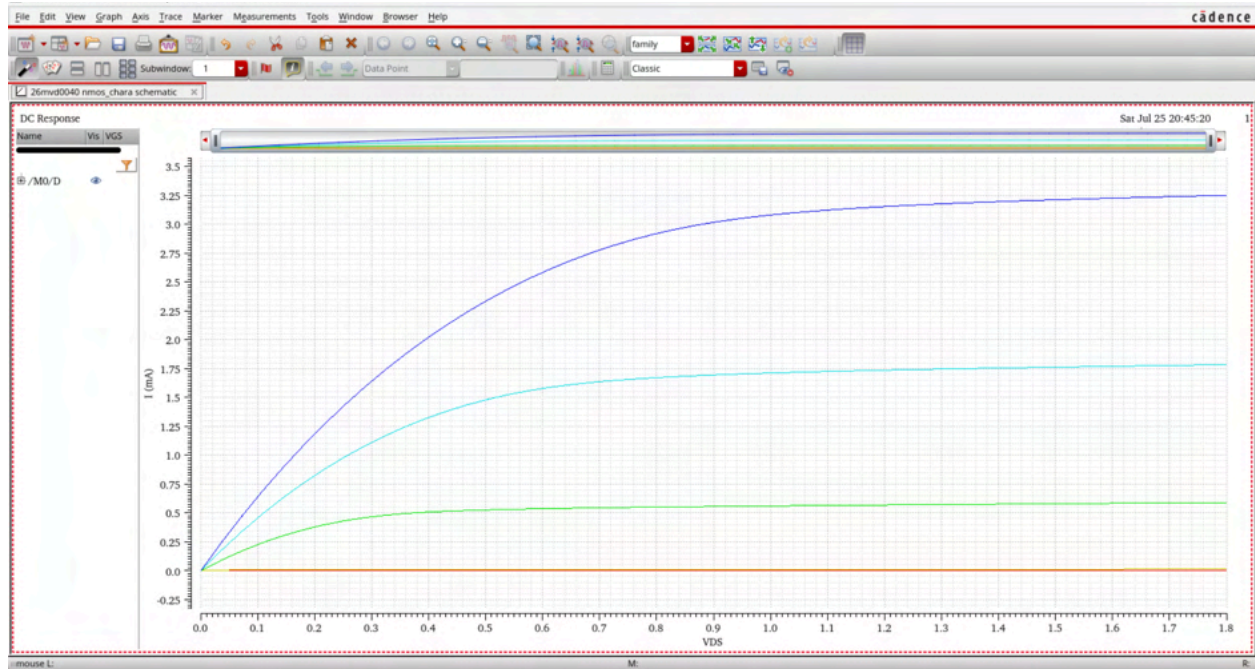


Figure 7: Output characteristics (ID) vs. (VDS) were obtained for different gate voltages.

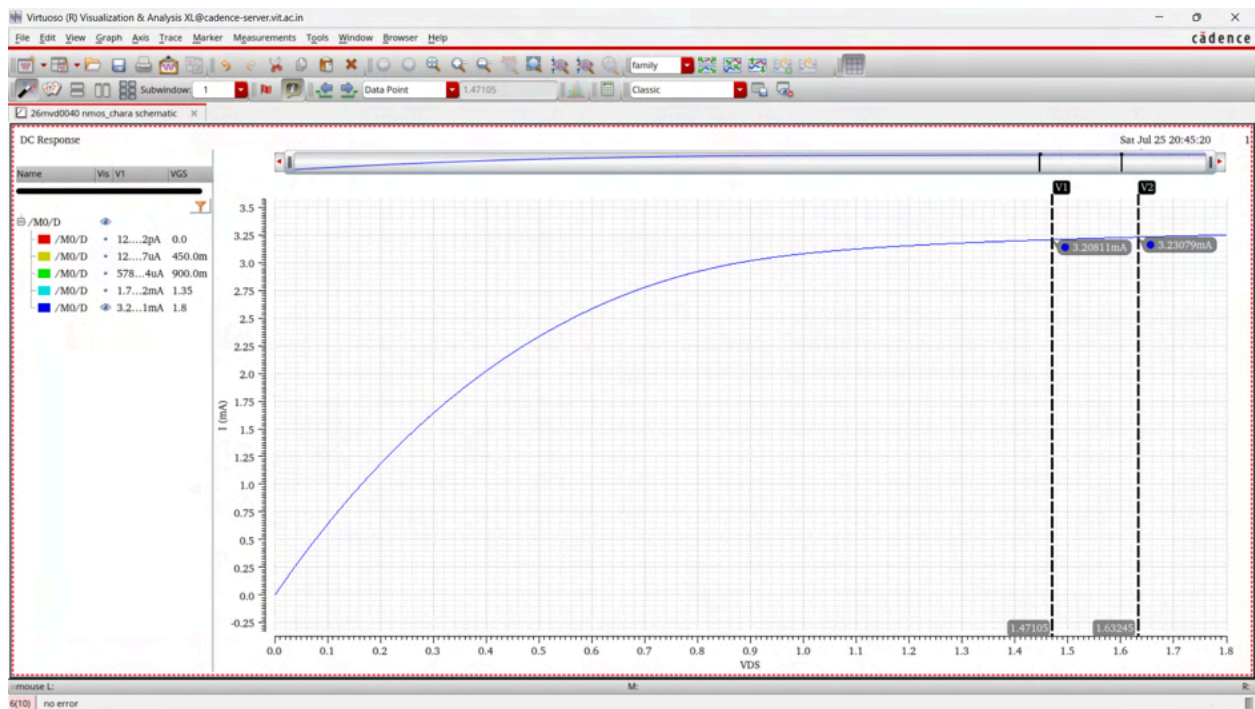


Figure 8: Output characteristics (ID) vs. (VDS) for λ calculation.

CALCULATIONS:

Channel length modulation λ is given by,

$$\lambda = \frac{(ID_2 - ID_1)}{(ID_1 V_{DS2} - ID_2 V_{DS1})}$$

$$V_{DS1} = 1.47105 \text{ V}$$

$$V_{DS2} = 1.63245 \text{ V}$$

$$ID_1 = 3.20811 \text{ mA}$$

$$ID_2 = 3.23079 \text{ mA}$$

$$\lambda = 0.0436 \text{ V}^{-1}$$

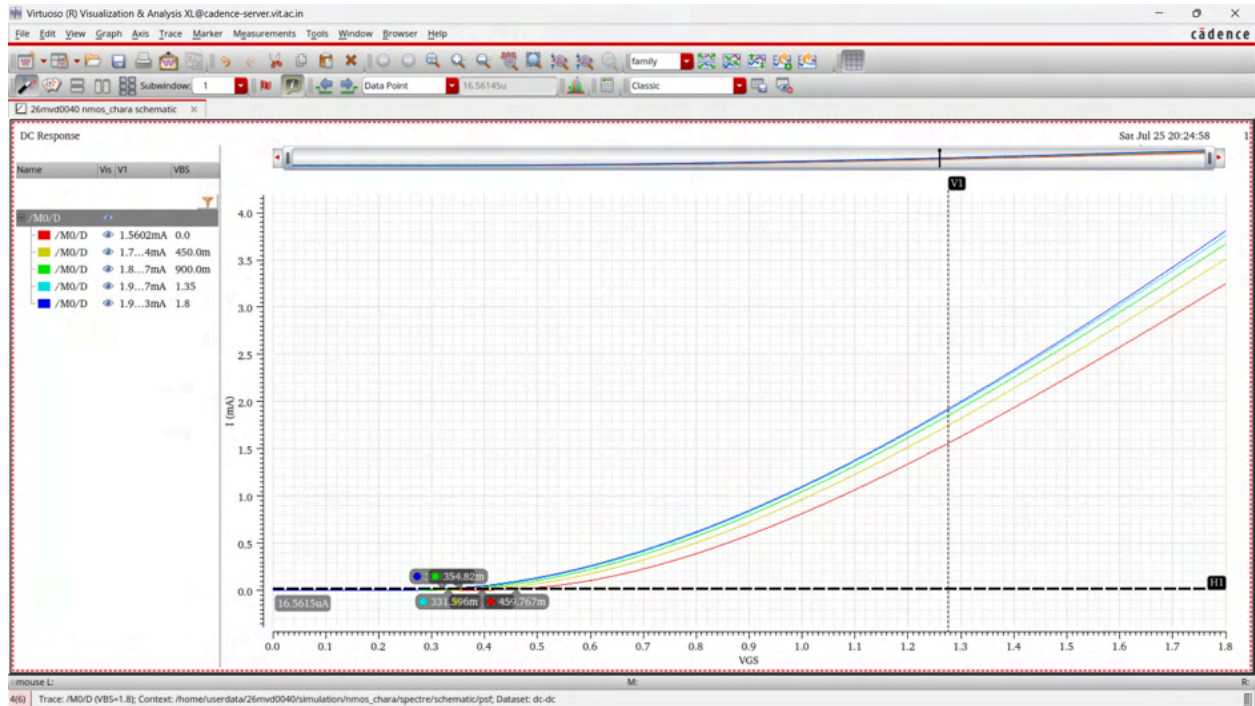


Figure 8: I_D V/s V_{GS} Characteristics of NMOS for Various Body Bias Voltages (V_{BS})

RESULT

The NMOS characteristics were successfully obtained using Cadence Virtuoso. The **ID V/s VDS** and **ID V/s VGS** characteristics were plotted. The **body effect** was verified by observing an increase in threshold voltage with increasing **VBS**. The **channel length modulation parameter (λ)** was calculated from the saturation region, and the simulation results agreed with the theoretical behavior of the NMOS transistor.

2. PMOS Characteristics

OBJECTIVE

- To obtain the Drain Current (ID) versus Gate-to-Source Voltage (VSG) characteristics of the MOS transistor.
- To obtain the Drain Current (ID) versus Drain-to-Source Voltage (VSD) characteristics for various values of VGS.
- To obtain the Drain Current (ID) versus Gate-to-Source Voltage (VSG) characteristics for various values of Source-to-Body Voltage (VBS) and study the body effect.

PROCEDURE

1. Open MobaXterm and create a new SSH session.
2. Connect to the Cadence server using the provided remote host, username, and password.
3. Launch the Cadence environment by entering `csd` and start Cadence Virtuoso.
4. Create a new library and attach it to the UMC-18 CMOS technology library.
5. Create a new schematic cell view.
6. Place a PMOS transistor (P_18_MM) from the UMC-18 CMOS library.
7. Place vdc voltage sources and gnd from the analogLib library.
8. Connect the circuit using the wire tool. Connect the PMOS source to VDD and the body (bulk) to the source.
9. Set the transistor dimensions:
 - Channel Length (L) = 500 nm
 - Channel Width (W) = 10 μ m
10. Define the voltage sources:
 - V0 = VSG (Gate-to-Source Voltage)
 - V1 = VSD (Source-to-Drain Voltage)
11. Save and check the schematic for errors.

12. Launch ADE L (Analog Design Environment).
13. Load the UMC-18 CMOS Spectre model library and select the TT process corner.
14. Select DC Analysis and enable Save DC Operating Point.
15. Copy the design variables from the cell view.
16. Perform DC sweep and parametric analysis to obtain:
 - (ID) vs. (VSG)
 - (ID) vs. (VSD) for various values of (VSG)
 - (ID) vs. (VSG) for various values of (VBS) (body bias)
17. Run the simulation, plot the required characteristics, and record the observations and results.

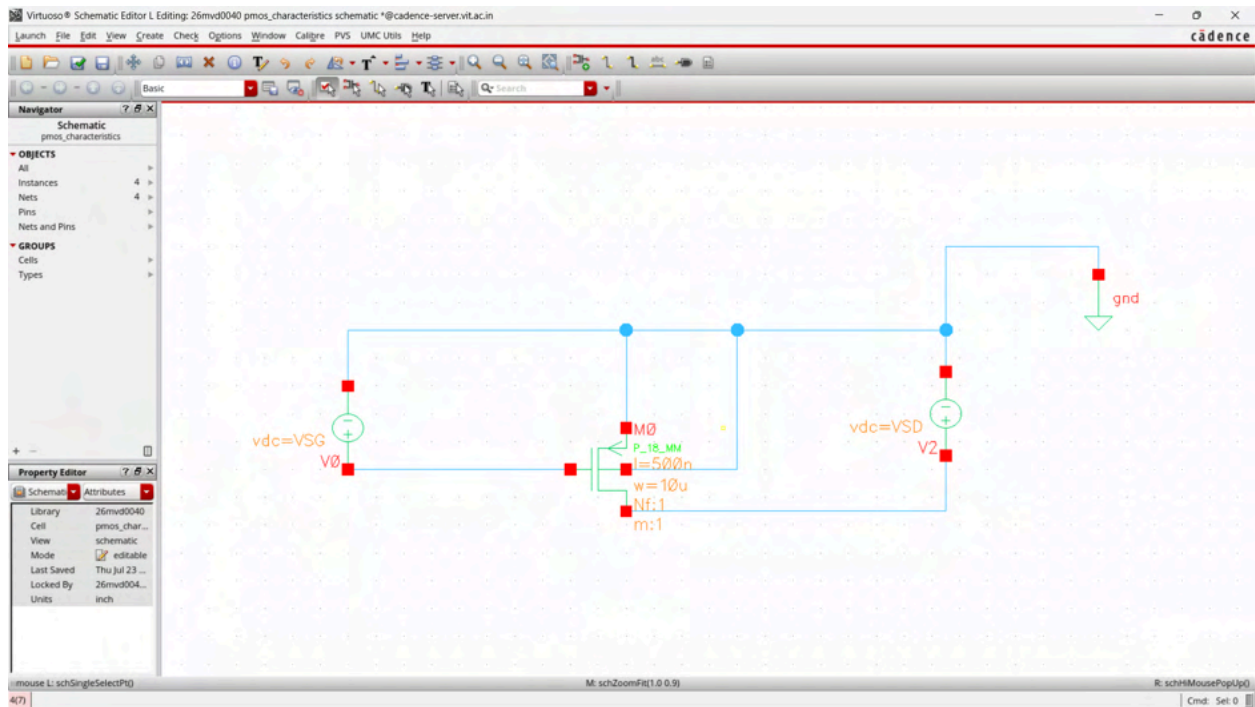


Figure 9: PMOS Characterization Circuit for ID V/s VSG and ID V/s VSD Analysis .

The circuit consists of a **PMOS transistor** configured to obtain its **output characteristics (ID-VSD)**. The **source** and **body (bulk)** terminals are connected together and grounded as the reference node, ensuring ($V_{SB}=0$). The **gate** terminal is connected to a variable DC voltage source (**V0**) that provides the **source-to-gate voltage (VSG)**, while the **drain** terminal is connected to another variable DC voltage source (**V2**) that provides the **source-to-drain voltage (VSD)**. During DC analysis, (VSD) is swept over the required range for different fixed values of (VSG), and the corresponding drain current (ID) is measured. The resulting characteristics are used to study the linear and saturation regions of PMOS operation and to determine the channel length modulation parameter (λ).

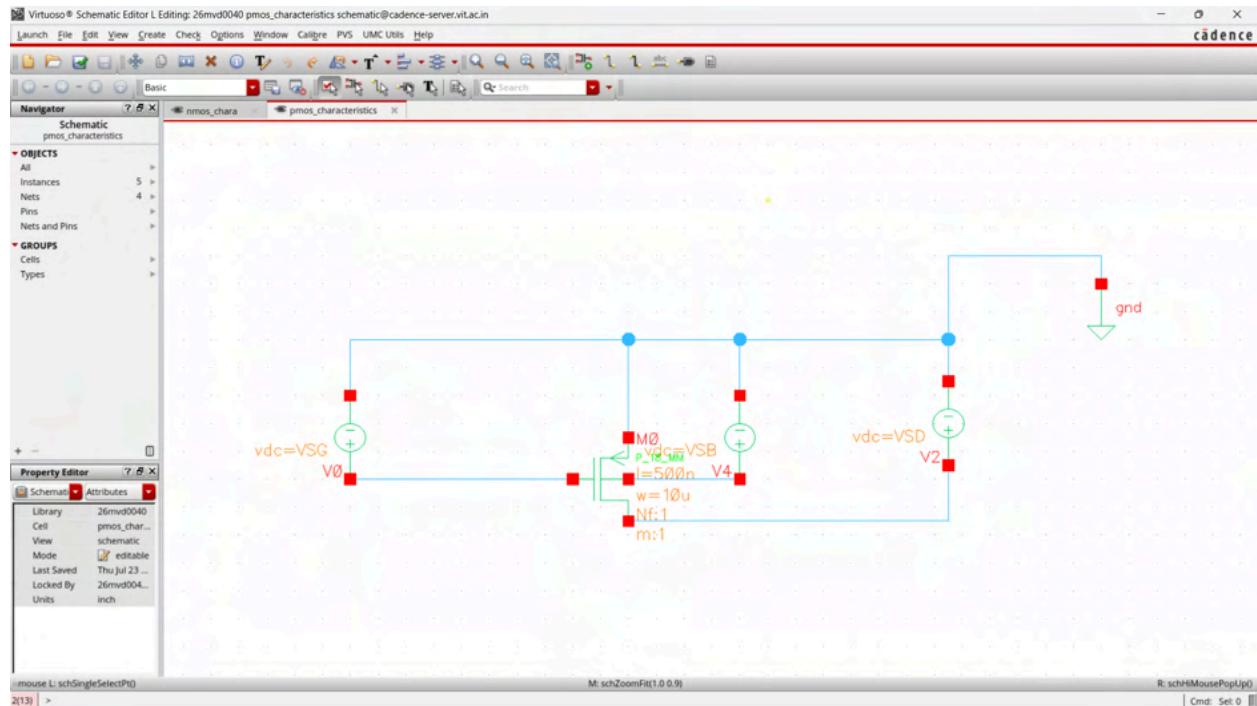


Figure 10: PMOS Test Circuit for Body Effect Analysis (ID) Vs (VSG) for Various (VSB)

The circuit uses a **PMOS transistor** to study the **body effect**. A variable DC source (**V0**) applies (**VSG**), **V2** applies (**VSD**), and **V4** provides the **body-to-source voltage (VSB)**. By sweeping (**VSG**) for different values of (**VSB**), the drain current is measured to obtain the transfer characteristics and observe the effect of body bias on the threshold voltage of the PMOS transistor.

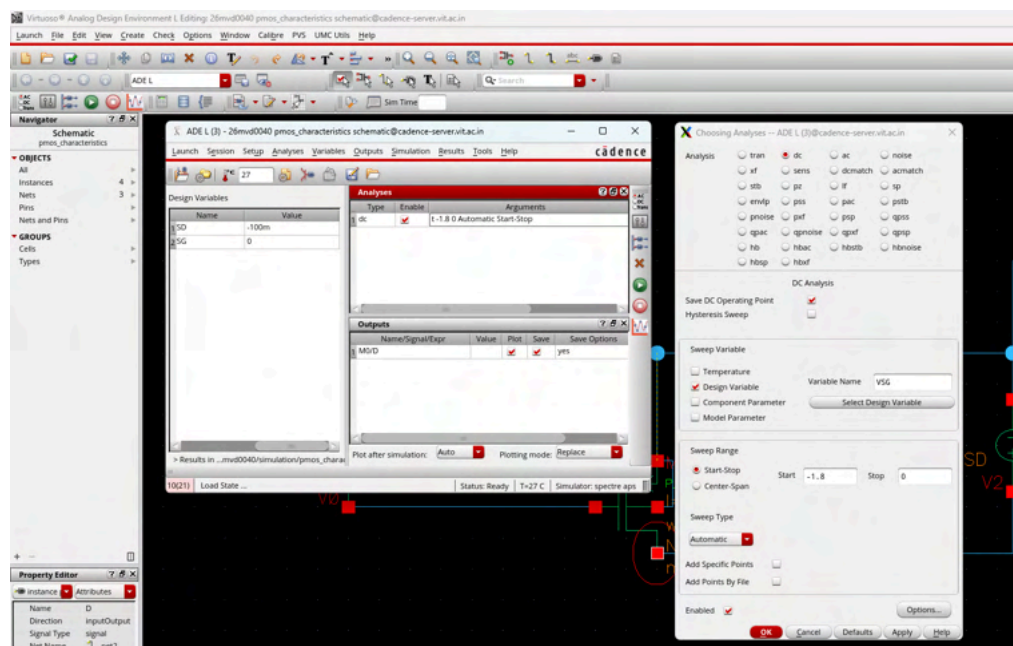


Figure 11: Configuration of DC Analysis for NMOS ID V/s VSG Simulation.

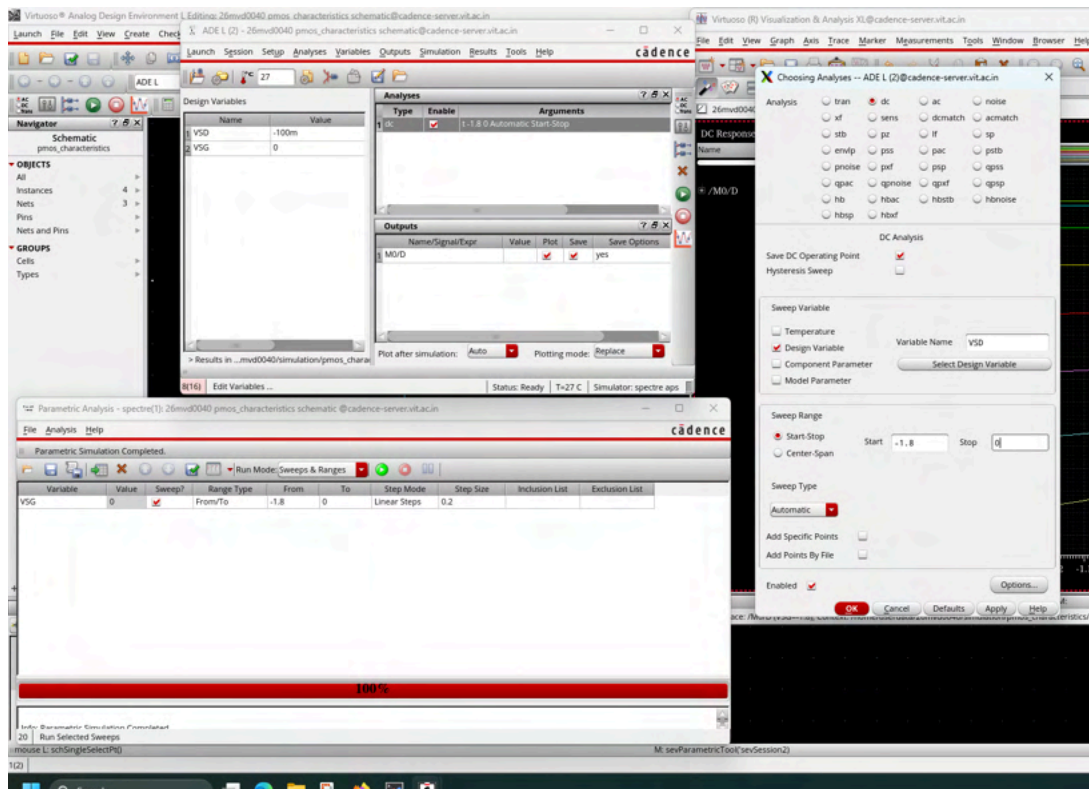


Figure 12: Configuration of DC Analysis for NMOS ID V/s VSD Simulation.

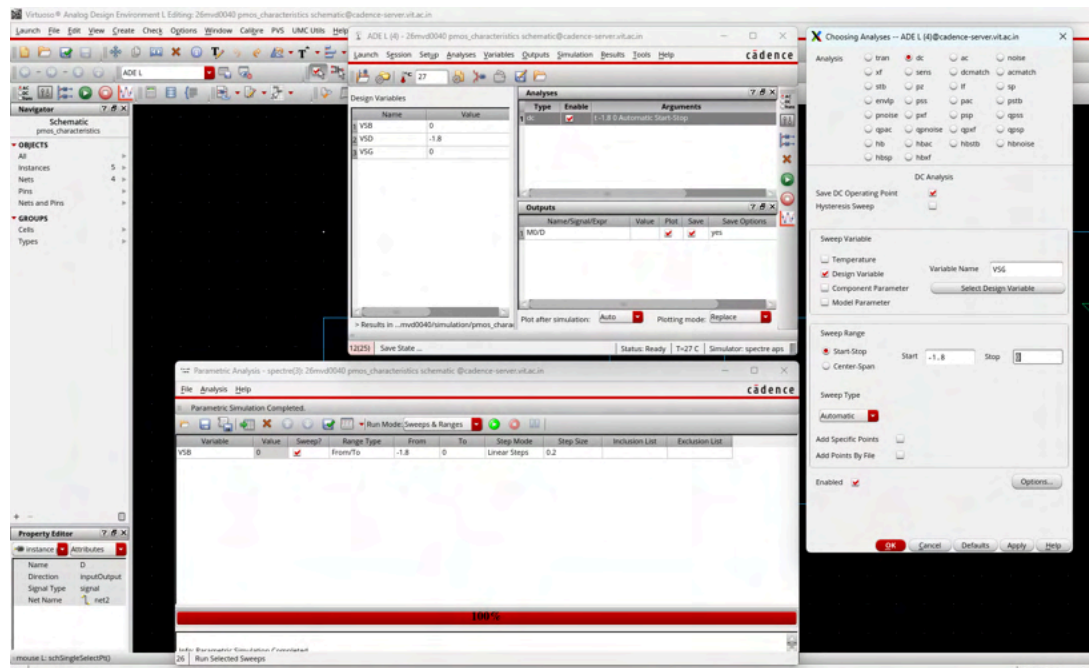


Figure 13: DC Parametric Sweep for ID V/s VSG Characteristics with Variable VSB.

OBSERVATIONS

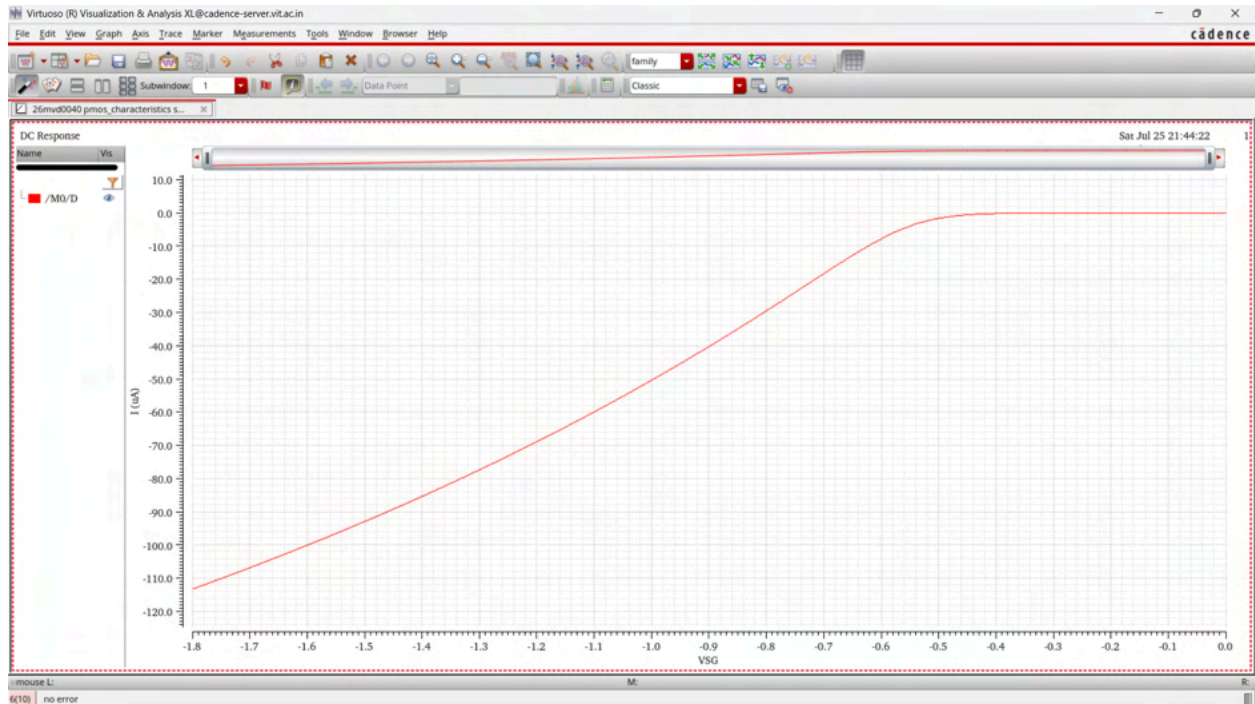


Figure 14: Transfer characteristics (I_D) vs. (VSG) were obtained

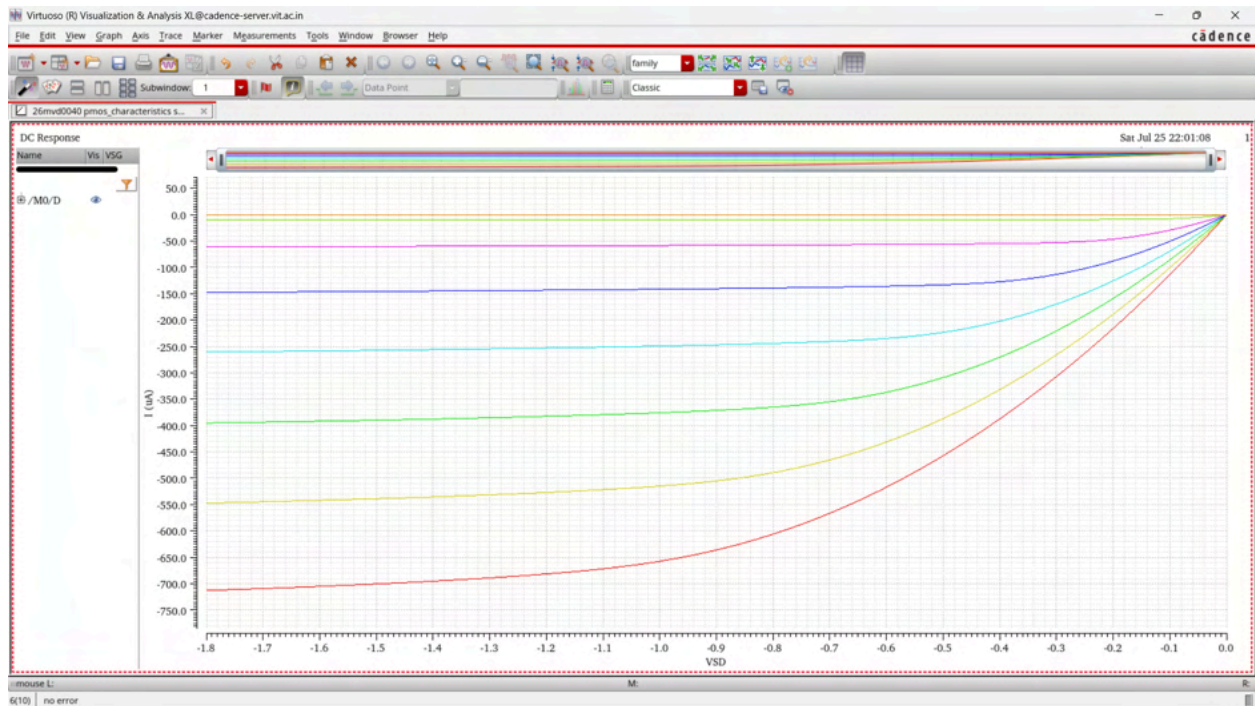


Figure 15: Output characteristics (I_D) vs. (VSD) were obtained for different gate voltages

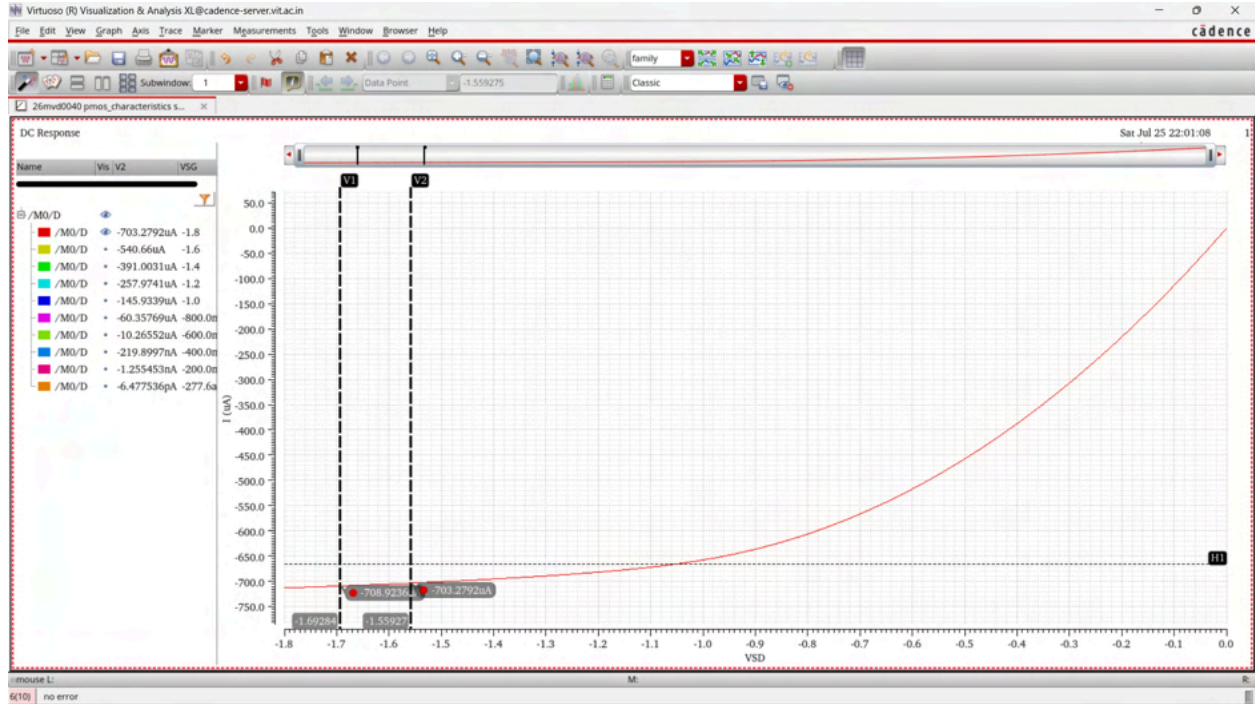


Figure 16: Output characteristics (I_D) vs. (V_{SD}) for λ calculation.

CALCULATIONS:

Channel length modulation λ is given by,

$$\lambda = \frac{(I_{D1} - I_{D2})}{(I_{D1} V_{SD2} - I_{D2} V_{SD1})}$$

$$V_{SD1} = -1.69284 \text{ V}$$

$$V_{SD2} = -1.55927 \text{ V}$$

$$I_{D1} = -708.9236 \mu A$$

$$I_{D2} = -703.2792 \mu A$$

$$\lambda = 0.0599 \text{ V}^{-1}$$

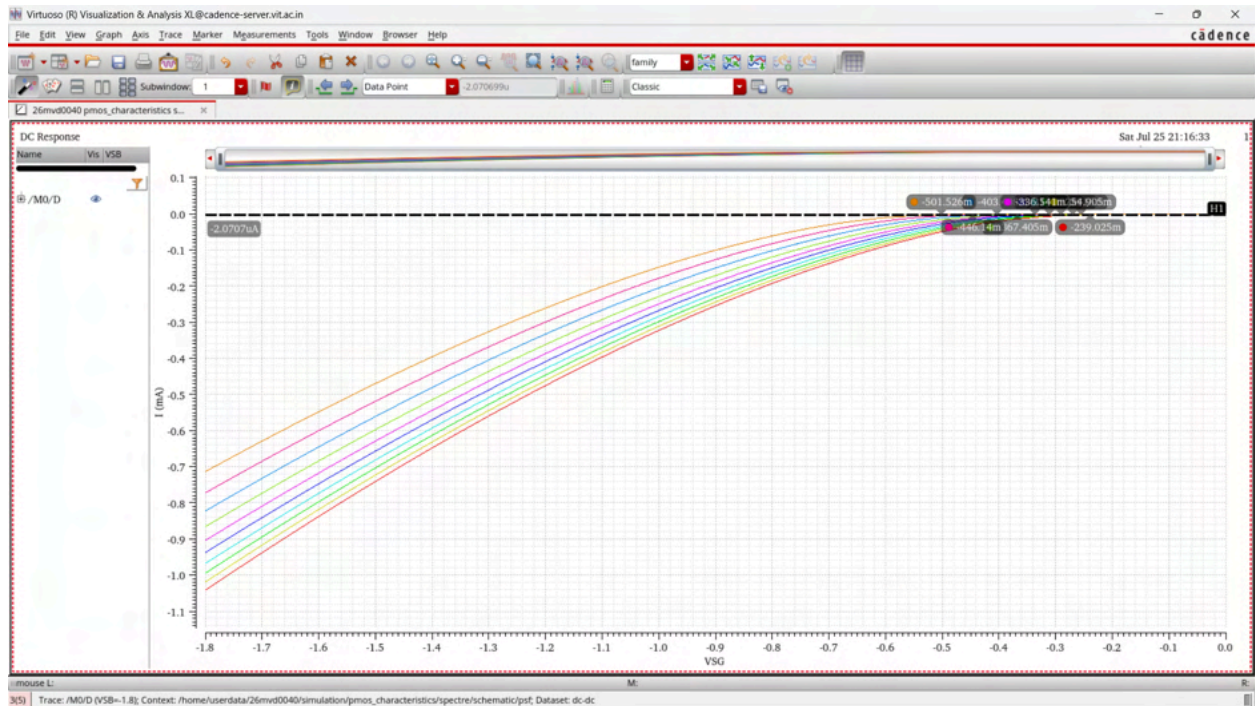


Figure 17: I_D V/s V_{SG} Characteristics of PMOS for Various Body Bias Voltages (V_{SB})

RESULT

The PMOS characteristics were successfully obtained using Cadence Virtuoso. The I_D V/s V_{SD} and I_D V/s V_{SG} characteristics were plotted. The **body effect** was verified by observing an increase in threshold voltage with increasing V_{SB} . The **channel length modulation parameter** (λ) was calculated from the saturation region, and the simulation results agreed with the theoretical behavior of the PMOS transistor.